Francisco Miguel Castro-Macías

Curriculum Vitae

Personal information

Full name	Francisco Miguel Castro Macías
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Nacionality	Spain
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Education

2023– **PhD. in Computer Science**, *Probabilistic machine learning methods for weakly supervised and inverse problems. Applications in medicine*, University of Granada, Granada.

2021–2022 **Msc. in Data Science and Computer Engineering**, University of Granada, Granada, GPA: 9.68/10.0.

2017–2021 **Degree in Mathematics**, University of Granada, Granada, GPA: 9.008/10.0.

2017–2021 **Degree in Computer Engineering**, University of Granada, Granada, GPA: 9.077/10.0.
Specialty in Computing and Intelligent Systems

Positions

Jan. 2023 – Jan. 2027 **FPU Predoctoral Researcher**, Department of Computer Science and Artificial Intelligence, University of Granada, Granada.

Jul. 2022 – Dec. 2022 **Project-funded researcher**, *Development and evaluation of methods for deblurring and color normalization of histological images*, Visual Information Processing (VIP) group, Department of Computer Science and Artificial Intelligence, University of Granada, Granada.

Internships

March 2024 – June 2024 **Northwestern University**, Department of Electrical and Computer Engineering, Northwestern University, Evanston, Illinous, United States.

I worked under the supervision of Prof. Aggelos K. Katsaggelos for three months as a visiting student.

Grants

Jan. 2023 – Jan. 2027 **Formación Profesorado Universitario (FPU)**, Department of Computer Science and Artificial Intelligence, University of Granada, Granada, Awarded by the Ministry of Universities of the Government of Spain.

This grant funded my PhD studies.

Aug. 2022 **Blas Cabrera Summer School**, Universidad Internacional Menéndez Pelayo, Awarded by the Ministry of Universities.

One-week stay at the Palacio de la Magdalena, Santander. Attendance at seminars and talks given by the country's most renowned researchers

Sep. 2020 – July 2021 **Research Initiation for Undergraduate Students**, *A federated learning model to predict COVID19 disease from chest x-rays images*, Department of Computer Science and Artificial Intelligence, University of Granada, Granada.

Extension of several deep learning models to the federated scenario. Theoretical and experimental study of various aggregators and data partitions. Under the supervision of Francisco Herrera Triguero and Siham Tabik Ouled Hrour.

2017 **English language immersion (Level C1)**, Universidad Internacional Menéndez Pelayo, Huesca.

July 2015 **Campus Científicos de Verano (Summer Science Campus)**, *TIC-TAC Robot*, Escuela Técnica Superior de Ingeniería, Departament d'Enginyeria Elèctrica, Electrònica i Automàtica, Universitat Rovira i Virgili, Tarragona.

Sep. 2014 – May 2015 **Secondary School Innovation Research Initiation Project in Andalusia**, *Stellar Clusters*, Consejo Superior de Investigaciones Científicas, University of Granada, Granada.

Awards

Nov. 2024 **Best Student Paper Award at 2024 IEEE International Conference on Image Processing (ICIP)**, *Bayesian Blind Image Deconvolution using a Hyperbolic-Secant prior*, Link to the news.

May 2022 **Ex aequo award for the best Final Degree Project of the Degree in Computer Engineering**, School of Computer Science and Telecommunications Engineering, University of Granada, Granada.

Research

Publications

2024 **F. M. Castro-Macías**, P. Morales-Álvarez, Y. Wu, R. Molina, A. K. Katsaggelos, *Sm: enhanced localization in Multiple Instance Learning for medical imaging classification*, 2024 Conference on Neural Information Processing Systems (NeurIPS).

2024 **F. M. Castro-Macías**, F. Pérez-Bueno, M. Vega, J. Mateos, R. Molina, A. K. Katsaggelos, *Bayesian Blind Image Deconvolution using a Hyperbolic-Secant prior*, 2024 IEEE International Conference on Image Processing (ICIP), **Best Student Paper Award**.

2024 **F. M. Castro-Macías**, F. Pérez-Bueno, M. Vega, J. Mateos, R. Molina, A. K. Katsaggelos, *Blind Color Deconvolution and Classification of Histological Images Using the Hyperbolic Secant Prior*, 2024 International Symposium on Biomedical Imaging (ISBI).

2024 **F. M. Castro-Macías**, P. Morales-Álvarez, Y. Wu, R. Molina, A. K. Katsaggelos, *Hyperbolic Secant representation of the logistic function: application to Probabilistic Multiple Instance Learning for CT intracranial hemorrhage detection*, Artifical Intelligence Journal.

2023 S. Yang, F. Pérez-Bueno, **F. M. Castro-Macías**, R. Molina, A. K. Katsaggelos, *BCD-Net: Stain Separation of Histological Images using Deep Variational Bayesian Blind Color Deconvolution*, Digital Signal Processing.

2023 Y. Wu, **F. M. Castro-Macías**, P. Morales-Álvarez, R. Molina, A. K. Katsaggelos, *Smooth Attention for Deep Multiple Instance Learning: Application to CT Intracranial Hemorrhage Detection*, 2023 International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI).

2023 S. Yang, F. Pérez-Bueno, **F. M. Castro-Macías**, R. Molina, A. K. Katsaggelos, *Deep Bayesian Blind Color Deconvolution of histological images*, 2023 IEEE International Conference on Image Processing (ICIP).

Training courses

August 2022 **OxML Summer School: Health Track**, Mathematical Institute, University of Oxford, Oxford.

April 2022 **Second training school Cloud Artificial Intelligence for Pathology (Clarify) project (EU funded)**, School of Computer Science and Telecommunications Engineering, University of Granada, Granada.

Teaching

Courses

2024 – 2025 **Development and Design of Information Systems**, Bachelor's Degree in Computer Engineering, University of Granada, Granada.

2023 – 2024 **Development and Design of Information Systems**, Bachelor's Degree in Computer Engineering, University of Granada, Granada.

2023 – 2024 **ICT in Healthcare and Research Methodology**, Bachelor's Degree in Nursing, University of Granada, Granada.

2022 – 2023 **Image feature extraction (assistant)**, Master's degree in Data Science and Computer Engineering, University of Granada, Granada.

Bachelor's and Master's Thesis

2023 – 2024 **Blind Image Deconvolution using Deep Learning**, Bachelor's degree in Computer Engineering, University of Granada, Granada.
Student: J. A. Torres Coca, Supervisors: R. Molina, F. M. Castro-Macías.

2023 – 2024 **Uncertainty Estimation in Blind Image Deconvolution using Bayesian Deep Image Prior**, Master's degree in Data Science and Computer Engineering, University of Granada, Granada.
Student: A. Pérez Lara, Supervisors: R. Molina, F. M. Castro-Macías.

2022 – 2023 **Deep Multiple Instance Learning. Application to Intracranial Hemorrhage Detection**, Master's degree in Data Science and Computer Engineering, University of Granada, Granada.
Student: E. Jiménez Mariscal, Supervisors: P. Morales-Álvarez, F. M. Castro-Macías.

Outreach

Sep. 2024 **The challenges of artificial intelligence in medicine**, European Researchers' Night, Granada.

Sep. 2023 **The wisdom of crowds: Are many people smarter than one expert?**, European Researchers' Night, Granada.

May 2022 **Speaker of the presentation "Artificial Intelligence from the inside"**, Academia El Futuro SLU, Granada.

May 2022 **Speaker of the presentation "That so-called Artificial Intelligence"**, Centro de Estudios Hermanos Naranjo, Granada.

Jan. 2022 **Speaker of the presentation "That so-called Artificial Intelligence"**, Centro de Estudios Hermanos Naranjo, Granada.

Languages

Spanish Mother tongue

English Fluent

Sport achievements (Weightlifting)

Bronze Trophy "Feria de Almería de Halterofilia", -61Kg, 2019, Almería

Gold Spanish University Weightlifting Championships, –61Kg, 2019, Murcia

Silver Senior Weightlifting Championship of Andalucía, -61Kg, 2019, Cádiz